WEEK 002

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Definition (Additive uniqueness)**.** *Let $\mathcal{S}$ be a subset of the integers, $\mathbb{Z}$. $\mathcal{S}$ is said to satisfy additive uniqueness if, whenever $n_k \in \mathcal{S}$ for $k = 1, 2, 3, 4$ and $n_1 + n_2 = n_3 + n_4$, then either $n_1 = n_3$ or $n_1 = n_4$.*

Remark. *This can be interpreted as saying that addition is injective on $\mathcal{S}$ when seen as an operation on unordered pairs.*

Let $\mathcal{S}$ be a subset of $\mathbb{Z}$ with $N \geq 1$ elements satisfying additive uniqueness. Define $f\colon \mathbb{C} \to \mathbb{C}$ by

$$f(x) := \sum_{n \in \mathcal{S}} e^{2\pi i n x}.$$

(a) *Evaluate the following:*

$$(1) \quad \int_0^1 |f(x)|^2 \,\mathrm{d}x \qquad (2) \quad \int_0^1 |f(x)|^4 \,\mathrm{d}x$$

$|f(x)|^2$ can be expressed as the product of $f(x)$ with it's complex conjugate:

$$\begin{aligned}
|f(x)|^2 &= f(x)\overline{f(x)} \\
&= \left(\sum_{m \in \mathcal{S}} e^{2\pi i m x}\right)\left(\sum_{n \in \mathcal{S}} e^{-2\pi i n x}\right) \\
&= \sum_{m \in \mathcal{S}} \sum_{n \in \mathcal{S}} e^{2\pi i (m-n) x} \\
&= \left(\sum_{n \in \mathcal{S}} 1\right) + \left(\sum_{\substack{n, m \in \mathcal{S}, \\ n < m}} 4 \times \frac{e^{i(2\pi(m-n)x)} + e^{-i(2\pi(m-n)x)}}{2}\right) \\
&= N + \sum_{\substack{n, m \in \mathcal{S}, \\ n < m}} 4\cos(2\pi(m-n)x)
\end{aligned}$$

Thus,

$$\begin{aligned}
\int_0^1 |f(x)|^2 \,\mathrm{d}x &= \int_0^1 \left(N + \sum_{\substack{n, m \in \mathcal{S}, \\ n < m}} 4\cos(2\pi(m-n)x)\right) \mathrm{d}x \\
&= N,
\end{aligned}$$

since every cosine term contributes 0 to the integral.

We can use a similar trick to evaluate the second integral by considering $|f(x)|^4$ as $f(x)^2\overline{f(x)^2}$. We first expand $f(x)^2$:

$$\begin{aligned}
f(x)^2 &= \left(\sum_{m\in\mathcal{S}} e^{2\pi imx}\right)\left(\sum_{m\in\mathcal{S}} e^{2\pi imx}\right) \\
&= \sum_{m\in\mathcal{S}}\sum_{n\in\mathcal{S}} e^{2\pi i(m+n)x} \\
&= \sum_{n\in\mathcal{S}} e^{2\pi i(2n)x} + 2\sum_{\substack{n,m\in\mathcal{S}\\ n<m}} e^{2\pi i(n+m)x}
\end{aligned}$$

Here, additive uniqueness tells us that each of the terms $2n$, and $m+n$ indexed in these ways are distinct. thus, $f(x)^2$ essentially takes the form of $f(x)+2g(x)$ where g is the corresponding function to f when replacing $\mathcal{S}$ with $\{n+m \mid n,m\in\mathcal{S}\}$.

Repeating the expansion of the absolute square as the product with the complex conjugate:

$$\begin{aligned}
|f(x)|^4 &= f(x)^2\overline{f(x)^2} \\
&= (f(x)+2g(x))\overline{(f(x)+2g(x))} \\
&= f(x)\overline{f(x)} + 4g(x)\overline{g(x)} + 2\left(f(x)\overline{g(x)} + g(x)\overline{f(x)}\right)
\end{aligned}$$

We know how $f(x)\overline{f(x)}$ expands and what it contributes to the integral: N, and since $g(x)\overline{g(x)}$ has the same structure, it contributes $\#\{n+m \mid n,m\in\mathcal{S}\} = \frac{N(N-1)}{2}$. Finally, the $f(x)\overline{g(x)} + g(x)\overline{f(x)}$ corresponds to more cosine terms with non-zero integer multiples of 2π periods in the expansion, so it contributes 0. So, together,

$$\int_0^1 |f(x)|^4\,\mathrm{d}x = N + 2N(N-1) = 2N^2 - N.$$

(b) *Prove that*

$$\int_0^1 |f(x)|\,\mathrm{d}x \geq \frac{1}{2\sqrt{N}}.$$

When does the equality hold?

By Cauchy-Schwarz, we have,

$$\begin{aligned}
\left(\int_0^1 |f(x)|^2\,\mathrm{d}x\right)^2 &\leq \left(\int_0^1 |f(x)|\,\mathrm{d}x\right)\left(\int_0^1 |f(x)|^3\,\mathrm{d}x\right) \\
N^2 &\leq \left(\int_0^1 |f(x)|\,\mathrm{d}x\right)\left(\int_0^1 |f(x)|^3\,\mathrm{d}x\right).
\end{aligned}$$

A second application of Cauchy-Schwarz, gives:

$$\begin{aligned}\left(\int_0^1 |f(x)|^3 \,\mathrm{d}x\right)^2 &\leq \left(\int_0^1 |f(x)|^2 \,\mathrm{d}x\right)\left(\int_0^1 |f(x)|^4 \,\mathrm{d}x\right)\\ \left(\int_0^1 |f(x)|^3 \,\mathrm{d}x\right)^2 &\leq N^2(2N-1)\\ \int_0^1 |f(x)|^3 \,\mathrm{d}x &\leq N\sqrt{2N-1}.\end{aligned}$$

Putting them together, we have:

$$\int_0^1 |f(x)| \,\mathrm{d}x \geq \frac{N^2}{N\sqrt{2N-1}} = \frac{N}{\sqrt{2N-1}}$$

For all $N \geq 1$, $\frac{N}{\sqrt{2N-1}} > \frac{1}{2\sqrt{N}}$ since rearranging this gives the equivalent inequality $4N^3 > 2N - 1$ which is clearly true for all such N, so

$$\int_0^1 |f(x)| \,\mathrm{d}x \geq \frac{N}{\sqrt{2N-1}} > \frac{1}{2\sqrt{N}}$$

This tells us that the equality never holds.